kcoil_ecoil_sweep

IMP + JAX/BlackJAX sampler benchmark

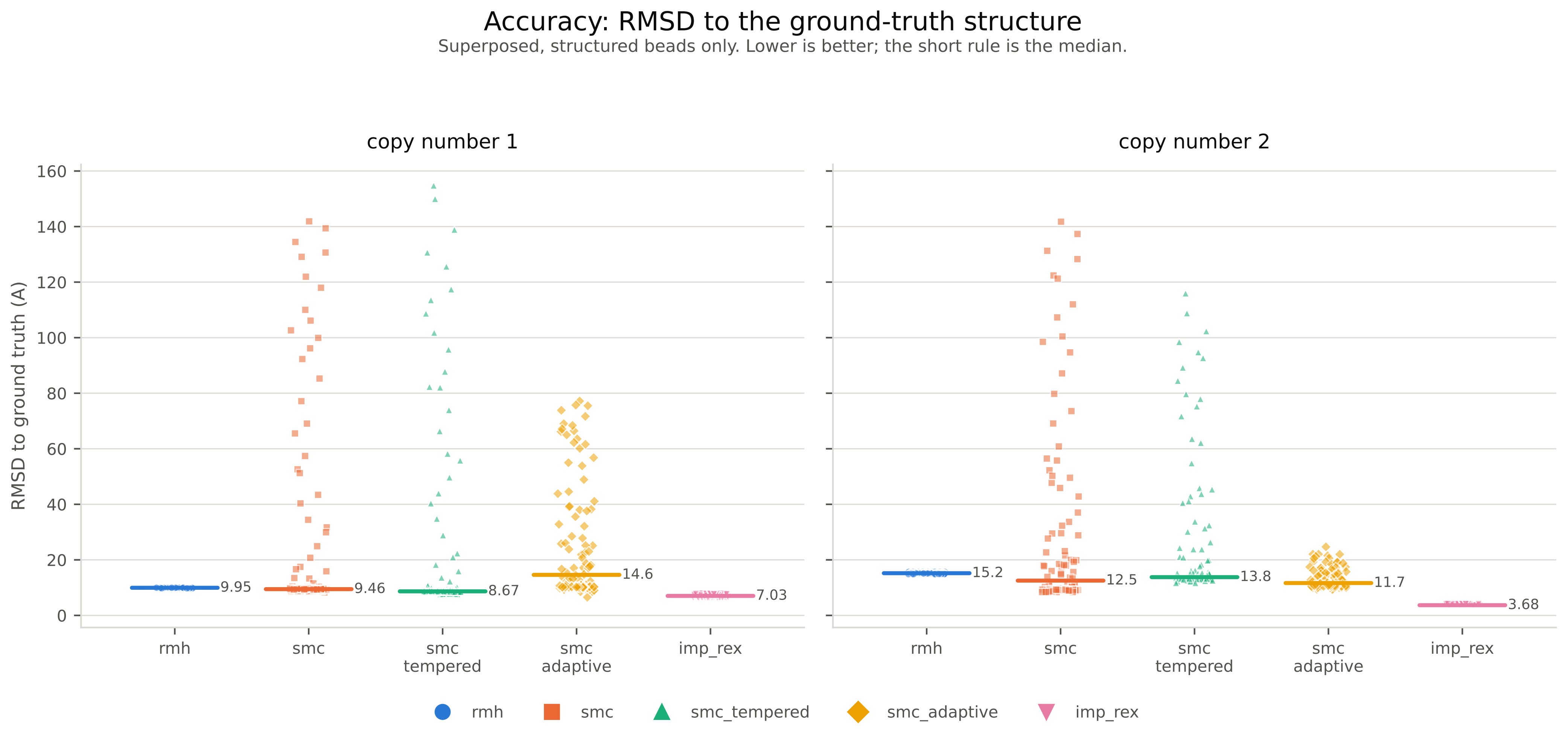
copy numbers	[1, 2]
samplers	rmh, smc, smc_tempered, smc_adaptive, imp_rex
JAX backend	cpu
prior	flat
restraints	{'top_n': 20, 'top_n_intra': 2, 'residue_types': 'KS', 'min_seq_sep': 3, 'force_constant': 1.0, 'wildca
scoring window	last 100 frames
satisfaction tol	2.0 A
cases run	10

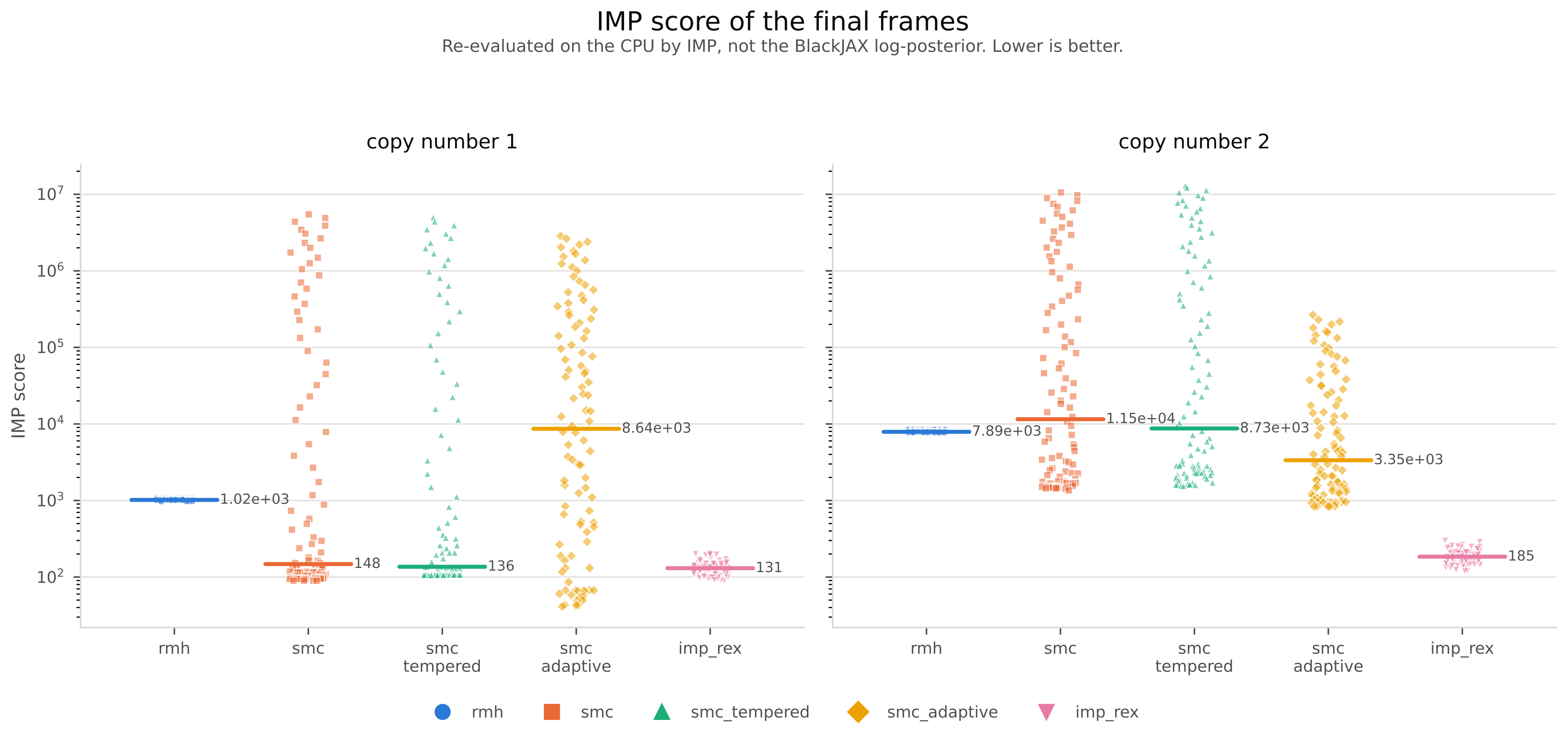
Scores on the score pages are computed by IMP on the CPU, by loading each frame back into an IMP model and evaluating the scoring function. They are NOT the BlackJAX log-posterior, which is -S(theta) + log p0(theta) -- a different quantity that would make BlackJAX and IMP samplers incomparable.

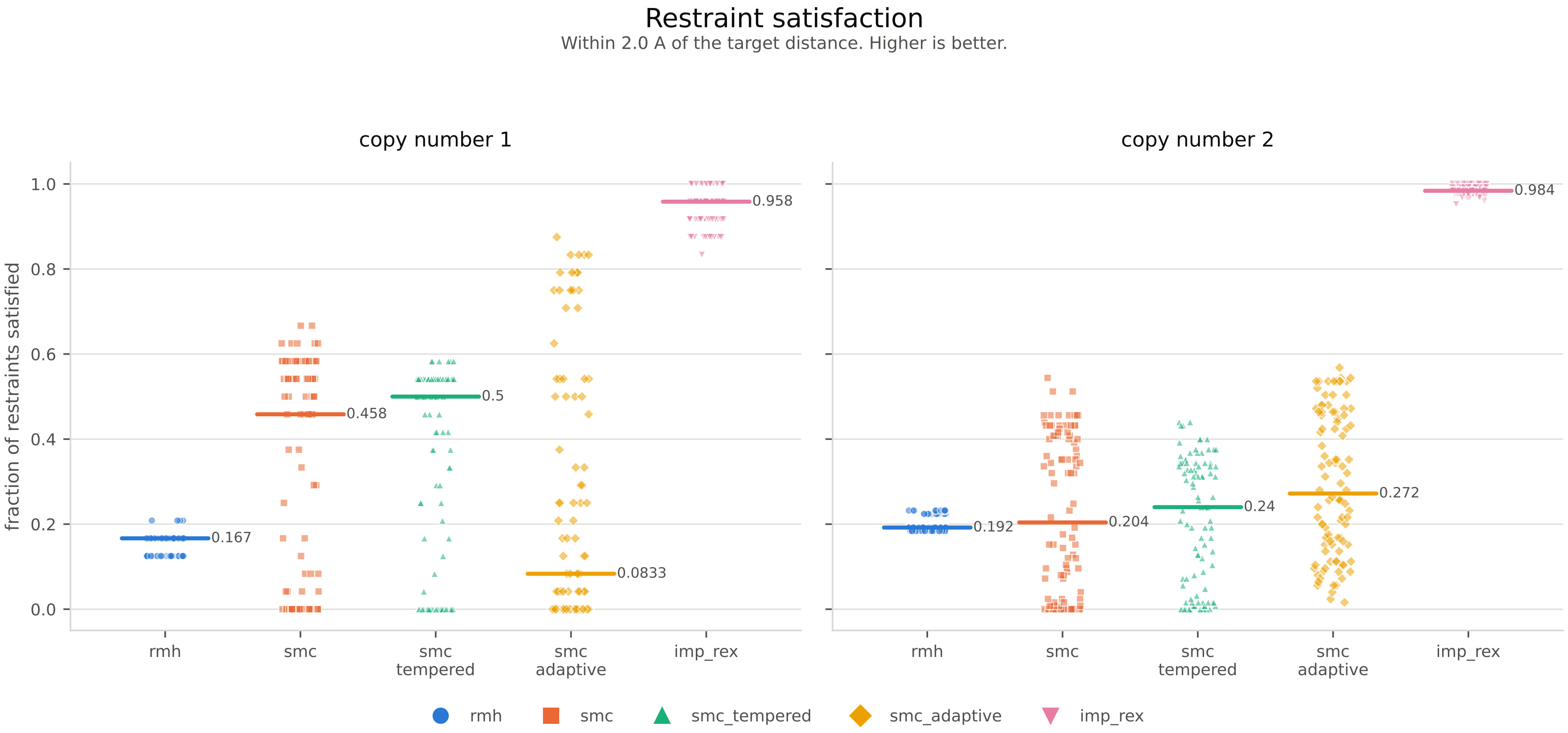
RMSD is superposed, over structured beads only, against the ground-truth structure. The flexible linker is excluded: it has no reference conformation to be right about.

Every point is one frame from the scoring window. The spread is the result, not noise.

NOTE: JAX ran on CPU for this sweep, so no CPU-vs-GPU comparison is possible from these numbers. Re-run on a GPU host to populate it.

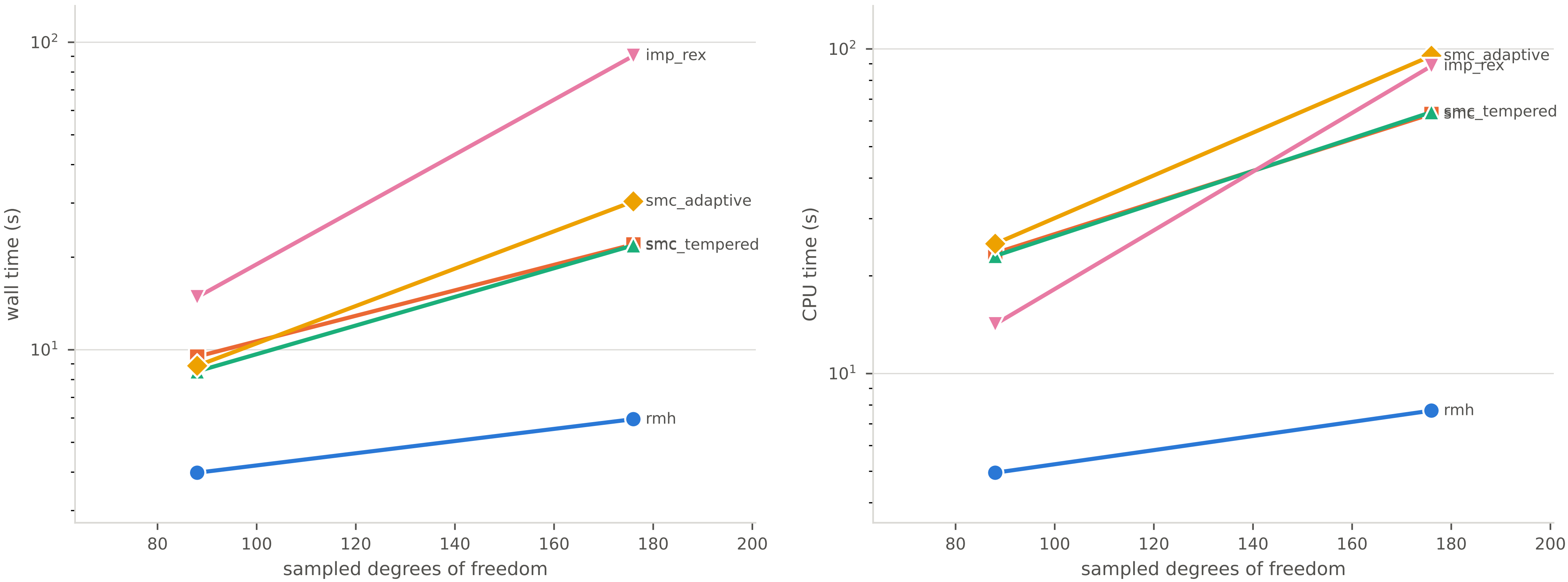

Accuracy: RMSD to the ground-truth structure
Superposed, structured beads only. Lower is better; the short rule is the median.
copy number 1
copy number 2
RMSD to ground truth (A)
0
20
40
60
80
100
120
140
160
9.95
9.46
8.67
14.6
7.03
15.2
12.5
13.8
11.7
3.68
rmh
smc
smc tempered
smc adaptive
imp_rex
rmh
smc
smc_tempered
smc_adaptive
imp_rex


IMP score of the final frames
Re-evaluated on the CPU by IMP, not the BlackJAX log-posterior. Lower is better.
copy number 1
copy number 2
IMP score
10^7
10^6
10^5
10^4
10^3
10^2
1.02e+03
148
136
8.64e+03
131
7.89e+03
1.15e+04
8.73e+03
3.35e+03
185
rmh
smc
smc tempered
smc adaptive
imp_rex
rmh
smc
smc_tempered
smc_adaptive
imp_rex


Restraint satisfaction
Within 2.0 A of the target distance. Higher is better.
copy number 1
copy number 2
fraction of restraints satisfied
1.0
0.8
0.6
0.4
0.2
0.0
0.958
0.458
0.5
0.167
0.0833
0.984
0.192
0.204
0.24
0.272
rmh
smc
smc tempered
smc adaptive
imp_rex
rmh
smc
smc_tempered
smc_adaptive
imp_rex

Scaling with system size

Restraint count is held to the same per-copy selection, so the growth is in degrees of freedom, not in data.



All results

copies	sampler	DOF	restr.	wall s	cpu s	RMSD min	RMSD med	IMP min	IMP med	satisfied	note
1	imp_rex	88	24	14.8	14.2	6.137	7.03	88.94	130.5	100%	frames=5000
1	rmh	88	24	4.0	4.9	9.738	9.948	978.6	1018	21%	acceptance=22.6%
1	smc	88	24	9.5	23.5	8.285	9.457	89.46	147.7	67%	particles=256 steps=10
1	smc_adaptive	88	24	8.9	25.1	6.534	14.6	41.32	8642	88%	particles=256 steps=10
1	smc_tempered	88	24	8.5	23.0	7.663	8.667	106.2	136.3	58%	particles=256 steps=10
2	imp_rex	176	125	90.4	88.6	3.228	3.68	118	184.6	100%	frames=5000
2	rmh	176	125	5.9	7.7	14.83	15.2	7760	7894	23%	acceptance=28.1%
2	smc	176	125	22.0	63.1	8.412	12.53	1351	1.153e+04	54%	particles=256 steps=10
2	smc_adaptive	176	125	30.4	95.2	9.321	11.66	834.7	3354	57%	particles=256 steps=15
2	smc_tempered	176	125	21.9	63.9	11.69	13.78	1548	8729	44%	particles=256 steps=10